WEEK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS

-S.Kaviya(E25AI119)

Scenario:

The Art WorkShop Company stores customer,product,order, and payment information in separate tables.The management wants to combine these related tables to understand customer purchases,order details,and payment status.

1.Implement INNER JOIN:

INNER JOIN – It returns only the matching records from both the tables.

TASKS:

1.Display customer details along with their orders.

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, O.ORDER_ID, O.TOTAL_AMOUNT FROM Customer C INNER JOIN Orders O ON C.C
USTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME                              ORDER_ID TOTAL_AMOUNT
---------------------------------------- ---------- ------------
Beema ponting                                  1001        70000
priya sakthi                                   1002        45000
Arun kumar                                     1003         3400
Anu Priya                                      1004        75000
Reena das                                      1011        60000
Beema ponting                                  1006        55000
priya sakthi                                   1007        67500
Arun kumar                                     1008        45000
Anu Priya                                      1009         3400
Reena das                                      1010         7500

10 rows selected.

SQL>
```

2.Display order details with payment information.

```
SQL> SELECT O.ORDER_ID, O.TOTAL_AMOUNT, O.ORDER_DATE, O.CUSTOMER_ID, P.PAYMENT_DATE, P.PAYMENT_MODE, P.PAYMENT_STATUS FROM Payment P J
OIN Orders O ON P.ORDER_ID = O.ORDER_ID;

  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
---------- ------------ --------- -------------------- ---------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
      1001        70000 20-SEP-26 C-1001               20-SEP-26
UPI                  Success

      1002        45000 21-SEP-26 C-1002               21-SEP-26
Card                 Success

      1003         3400 22-SEP-26 C-1003               22-SEP-26
Cash                 Success


  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
---------- ------------ --------- -------------------- ---------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
      1004        75000 23-SEP-26 C-1004               23-SEP-26
UPI                  Pending

      1011        60000 30-SEP-26 C-1005               30-SEP-26
UPI                  Success

      1006        55000 25-SEP-26 C-1001               25-SEP-26
Card                 Success


  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
```

```
  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
---------- ------------ --------- -------------------- ---------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
      1004        75000 23-SEP-26 C-1004               23-SEP-26
UPI                  Pending

      1011        60000 30-SEP-26 C-1005               30-SEP-26
UPI                  Success

      1006        55000 25-SEP-26 C-1001               25-SEP-26
Card                 Success


  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID          PAYMENT_D
---------- ------------ --------- -------------------- ---------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
      1007        67500 26-SEP-26 C-1002               26-SEP-26
UPI                  Pending

      1008        45000 27-SEP-26 C-1003               27-SEP-26
Card                 Success

      1009         3400 28-SEP-26 C-1004               28-SEP-26
Cash                 Success


9 rows selected.
```

3.Retrieve products purchased by customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      P.PRODUCT_ID,
  6      P.PRODUCT_NAME,
  7      OD.QUANTITY,
  8      OD.PRICE,
  9      OD.QUANTITY * OD.PRICE AS TOTAL_AMOUNT
 10  FROM CUSTOMER C
 11  JOIN ORDERS O
 12      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  JOIN ORDER_DETAILS OD
 14      ON O.ORDER_ID = OD.ORDER_ID
 15  JOIN PRODUCT P
 16      ON OD.PRODUCT_ID = P.PRODUCT_ID
 17  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID          CUSTOMER_NAME                                ORDER_ID
-------------------- -------------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1001               Beema ponting                                 1001
P1001
Laptops
         1      67500        67500
```

```
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1001                Beema ponting                                    1001
P1001
Laptops
         1      67500         67500


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1001                Beema ponting                                    1006
P1006
Tiles
         1      55000         55000


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1002                priya sakthi                                     1002
P1002
Croptops and shirts
         1      45000         45000
```

```
C-1002                priya sakthi                                     1007
P1001
Laptops
         1      67500         67500


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1003                Arun kumar                                       1003
P1003
Fish tank
         1       3400          3400


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1003                Arun kumar                                       1008
P1002
Croptops and shirts
         1      45000         45000


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1004                Anu Priya                                        1004
P1004
Television
         1      75000         75000


CUSTOMER_ID           CUSTOMER_NAME                                 ORDER_ID
```

```
CUSTOMER_ID          CUSTOMER_NAME                              ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1004               Anu Priya                                    1009
P1003
Fish tank
         1       3400         3400


CUSTOMER_ID          CUSTOMER_NAME                              ORDER_ID
-------------------- ---------------------------------------- ----------
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
C-1005               Reena das                                    1010
P1005
Unruled notes
         1       7500         7500


9 rows selected.
```

2.Implement LEFT JOIN:

LEFT JOIN – It returns all records from the left table and matching records from the right table.

TASKS:

1. Display all customers including customers who have not placed orders.

```
SQL> SELECT
  2       C.CUSTOMER_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       C.EMAIL,
  5       O.ORDER_ID AS ORDER_ID,
  6       O.ORDER_DATE AS ORDER_DATE,
  7       O.TOTAL_AMOUNT AS TOTAL_AMOUNT,
  8       O.ORDER_STATUS AS ORDER_STATUS
  9  FROM CUSTOMER C
 10  LEFT JOIN ORDERS O
 11       ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C-1001               Beema ponting
beema200@edss.com                  1001 20-SEP-26         70000
Pending

C-1001               Beema ponting
beema200@edss.com                  1006 25-SEP-26         55000
Delivered

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------

C-1002               priya sakthi
priyasakthi@edss.com               1002 21-SEP-26         45000
Shipped

C-1002               priya sakthi
priyasakthi@edss.com               1007 26-SEP-26         67500

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
Pending

C-1003               Arun kumar
```

```
C-1003               Arun kumar
arun200@edss.com                   1003 22-SEP-26          3400
Delivered

C-1003               Arun kumar

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
arun200@edss.com                   1008 27-SEP-26         45000
Shipped

C-1004               Anu Priya
anupriya200@edss.com               1004 23-SEP-26         75000
Pending


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C-1004               Anu Priya
anupriya200@edss.com               1009 28-SEP-26          3400
Delivered

C-1005               Reena das
reenadas200@edss.com               1011 30-SEP-26         60000
Pending

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------

C-1005               Reena das
reenadas200@edss.com               1010 29-SEP-26          7500
Pending

C-1006               Kaviya Suresh
kaviya106@gmail.com

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
```

```
C-1008                Divya Rani

CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                              ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
divya108@gmail.com


C-1009                Sanjay Raj
sanjay109@gmail.com



CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                              ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C-1010                Priya Devi
priya110@gmail.com


C-1011                Vijay Kumar
vijay111@gmail.com


CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                              ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------

C-1012                Meena Lakshmi
meena112@gmail.com


C-1013                Arjun Raj
arjun113@gmail.com

CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                              ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
```

```
C-1014                Sneha Rani
sneha114@gmail.com

C-1015                Nithin Kumar

CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                              ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
nithin115@gmail.com


20 rows selected.
```

2. Find customers without any purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      C.EMAIL,
  5      C.PHONE
  6  FROM CUSTOMER C
  7  LEFT JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  WHERE O.ORDER_ID IS NULL
 10  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                                    PHONE
---------------------------------------- ----------
C-1006               Kaviya Suresh
kaviya106@gmail.com                      9876543210

C-1007               Rahul Kumar
rahul107@gmail.com                       9876543211

C-1008               Divya Rani
divya108@gmail.com                       9876543212


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                                    PHONE
---------------------------------------- ----------
C-1009               Sanjay Raj
sanjay109@gmail.com                      9876543213

C-1010               Priya Devi
priya110@gmail.com                       9876543214

C-1011               Vijay Kumar
vijay111@gmail.com                       9876543215


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                                    PHONE
---------------------------------------- ----------
C-1012               Meena Lakshmi
meena112@gmail.com                       9876543216

C-1013               Arjun Raj
arjun113@gmail.com                       9876543217
```

```
C-1013               Arjun Raj
arjun113@gmail.com                       9876543217

C-1014               Sneha Rani
sneha114@gmail.com                       9876543218


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                                    PHONE
---------------------------------------- ----------
C-1015               Nithin Kumar
nithin115@gmail.com                      9876543219


10 rows selected.
```

3. Display all products including products with no sales.

```
SQL> SELECT
  2       P.PRODUCT_ID,
  3       P.PRODUCT_NAME,
  4       P.PRICE
  5  FROM PRODUCT P
  6  LEFT JOIN ORDER_DETAILS OD
  7       ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  WHERE OD.PRODUCT_ID IS NULL
  9  ORDER BY P.PRODUCT_ID;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P1007
Headphones
      2500

P1008
Keyboard
      1500

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

P1009
T-Shirts
       800

P1010
Jeans

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1800

P1011
Notebooks
       100

P1012

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Pens
        20

P1013
Mixer Grinder
      3500


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P1014
Mobile Charger
       600

P1015
Rice Bag
      1200

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------


9 rows selected.
```

3.Implement RIGHT JOIN :

RIGHT JOIN – It returns all records from the right table and matching records from the left table.

TASKS:

1. Display all orders with customer information.

```
SQL> SELECT ORDERS.*, CUSTOMER.*
  2  FROM CUSTOMER
  3  RIGHT JOIN ORDERS
  4  ON CUSTOMER.CUSTOMER_ID = ORDERS.CUSTOMER_ID;

  ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
      1006 C-1001                25-SEP-26        55000 Delivered
C-1001               Beema                ponting
beema200@edss.com               9234567856 14-FEB-23

      1001 C-1001                20-SEP-26        70000 Pending
C-1001               Beema                ponting
beema200@edss.com               9234567856 14-FEB-23

  ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------

      1007 C-1002                26-SEP-26        67500 Pending
C-1002               priya                sakthi
priyasakthi@edss.com            9856734456 23-SEP-24

      1002 C-1002                21-SEP-26        45000 Shipped
C-1002               priya                sakthi

  ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
priyasakthi@edss.com            9856734456 23-SEP-24
```

```
      1002 C-1002               21-SEP-26        45000 Shipped
C-1002               priya                sakthi

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
priyasakthi@edss.com            9856734456 23-SEP-24

      1008 C-1003               27-SEP-26        45000 Shipped
C-1003               Arun                 kumar
arun200@edss.com                9876543212 15-OCT-24

      1003 C-1003               22-SEP-26         3400 Delivered

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
C-1003               Arun                 kumar
arun200@edss.com                9876543212 15-OCT-24

      1009 C-1004               28-SEP-26         3400 Delivered
C-1004               Anu                  Priya
anupriya200@edss.com            8765943256 23-MAY-25


  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
      1004 C-1004               23-SEP-26        75000 Pending
C-1004               Anu                  Priya
anupriya200@edss.com            8765943256 23-MAY-25
```

```
      1010 C-1005               29-SEP-26         7500 Pending
C-1005               Reena                das
reenadas200@edss.com            9087654567 30-APR-22

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------

      1011 C-1005               30-SEP-26        60000 Pending
C-1005               Reena                das
reenadas200@edss.com            9087654567 30-APR-22


10 rows selected.
```

2. Find orders where customer details are missing.

```
SQL> SELECT ORDERS.*, CUSTOMER.*
  2  FROM CUSTOMER
  3  RIGHT JOIN ORDERS
  4  ON CUSTOMER.CUSTOMER_ID = ORDERS.CUSTOMER_ID
  5  WHERE CUSTOMER.CUSTOMER_ID IS NULL;

no rows selected
```

3. Display all payment records with order details.

```
SQL> SELECT PAYMENT.*, ORDERS.*
  2  FROM ORDERS
  3  RIGHT JOIN PAYMENT
  4  ON ORDERS.ORDER_ID = PAYMENT.ORDER_ID
  5  ORDER BY PAYMENT.PAYMENT_DATE;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------
         1       1001 20-SEP-26 UPI                  Success
             70000        1001 C-1001               20-SEP-26         70000
Pending

         2       1002 21-SEP-26 Card                 Success
             45000        1002 C-1002               21-SEP-26         45000
Shipped

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------

         3       1003 22-SEP-26 Cash                 Success
              3400        1003 C-1003               22-SEP-26          3400
Delivered

         4       1004 23-SEP-26 UPI                  Pending
             75000        1004 C-1004               23-SEP-26         75000

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------
Pending

         5       1006 25-SEP-26 Card                 Success
             55000        1006 C-1001               25-SEP-26         55000
Delivered

         6       1007 26-SEP-26 UPI                  Pending

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------
             67500        1007 C-1002               26-SEP-26         67500
Pending

         7       1008 27-SEP-26 Card                 Success
             45000        1008 C-1003               27-SEP-26         45000
Shipped


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------
         8       1009 28-SEP-26 Cash                 Success
              3400        1009 C-1004               28-SEP-26          3400
Delivered

        10       1011 30-SEP-26 UPI                  Success
             60000        1011 C-1005               30-SEP-26         60000
Pending

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT
------------------ ---------- -------------------- --------- ------------
ORDER_STATUS
--------------------


9 rows selected.
```

4. Retrieve Complete Order Details :

TASKS: Generate a complete order report by combining:

- Customer Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```
SQL> SELECT
  2      CUSTOMER.*,
  3      P.PRODUCT_NAME,
  4      OD.QUANTITY,
  5      O.ORDER_DATE,
  6      O.TOTAL_AMOUNT,
  7      PM.PAYMENT_STATUS
  8  FROM CUSTOMER
  9  JOIN ORDERS O
 10      ON CUSTOMER.CUSTOMER_ID = O.CUSTOMER_ID
 11  JOIN ORDER_DETAILS OD
 12      ON O.ORDER_ID = OD.ORDER_ID
 13  JOIN PRODUCT P
 14      ON OD.PRODUCT_ID = P.PRODUCT_ID
 15  LEFT JOIN PAYMENT PM
 16      ON O.ORDER_ID = PM.ORDER_ID
 17  ORDER BY O.ORDER_DATE, CUSTOMER.CUSTOMER_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1001               Beema                ponting
beema200@edss.com              9234567856 14-FEB-23
Laptops
         1 20-SEP-26        70000 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1002                priya                sakthi
priyasakthi@edss.com           9856734456 23-SEP-24
Croptops and shirts
         1 21-SEP-26        45000 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
---------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1003                Arun                 kumar
arun200@edss.com               9876543212 15-OCT-24
Fish tank
         1 22-SEP-26         3400 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
---------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1004                Anu                  Priya
anupriya200@edss.com           8765943256 23-MAY-25
Television
         1 23-SEP-26        75000 Pending


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
---------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
```

```
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1001                Beema                ponting
beema200@edss.com               9234567856 14-FEB-23
Tiles
         1 25-SEP-26        55000 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1002                priya                sakthi
priyasakthi@edss.com            9856734456 23-SEP-24
Laptops
         1 26-SEP-26        67500 Pending


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1003                Arun                 kumar
arun200@edss.com                9876543212 15-OCT-24
Croptops and shirts
         1 27-SEP-26        45000 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1004                Anu                  Priya
```

```
C-1004                Anu                  Priya
anupriya200@edss.com            8765943256 23-MAY-25
Fish tank
         1 28-SEP-26         3400 Success


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
C-1005                Reena                das
reenadas200@edss.com            9087654567 30-APR-22
Unruled notes
         1 29-SEP-26         7500


9 rows selected.
```

5.Display Customer Purchase History :

TASKS: Write queries to find:

1. All products purchased by a customer.

```
SQL> SELECT
  2      CUSTOMER.*,
  3      O.ORDER_ID AS ORDER_ID,
  4      O.ORDER_DATE AS ORDER_DATE,
  5      P.PRODUCT_NAME,
  6      OD.QUANTITY,
  7      OD.PRICE
  8  FROM CUSTOMER
  9  JOIN ORDERS O
 10      ON CUSTOMER.CUSTOMER_ID = O.CUSTOMER_ID
 11  JOIN ORDER_DETAILS OD
 12      ON O.ORDER_ID = OD.ORDER_ID
 13  JOIN PRODUCT P
 14      ON OD.PRODUCT_ID = P.PRODUCT_ID
 15  WHERE CUSTOMER.CUSTOMER_ID = 'C-1001'
 16  ORDER BY O.ORDER_DATE;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CREATED_A   ORDER_ID ORDER_DAT
------------------------------ ---------- --------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
C-1001               Beema                ponting
beema200@edss.com             9234567856 14-FEB-23       1001 20-SEP-26
Laptops
         1      67500


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                          PHONE CREATED_A   ORDER_ID ORDER_DAT
------------------------------ ---------- --------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
C-1001               Beema                ponting
beema200@edss.com             9234567856 14-FEB-23       1006 25-SEP-26
Tiles
         1      55000
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2       C.CUSTOMER_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       SUM(O.TOTAL_AMOUNT) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9       C.CUSTOMER_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
C-1001               Beema ponting
             125000

C-1002               priya sakthi
             112500

C-1004               Anu Priya
              78400


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
C-1005               Reena das
              67500

C-1003               Arun kumar
              48400
```

3. Number of orders placed by each customer.

```
SQL> SELECT
  2       C.CUSTOMER_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7       ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9       C.CUSTOMER_ID,
 10       C.FIRST_NAME,
 11       C.LAST_NAME
 12  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_ID          CUSTOMER_NAME                            NUMBER_OF_ORDERS
-------------------- ---------------------------------------- ----------------
C-1007               Rahul Kumar                                             0
C-1008               Divya Rani                                              0
C-1014               Sneha Rani                                              0
C-1011               Vijay Kumar                                             0
C-1010               Priya Devi                                              0
C-1012               Meena Lakshmi                                           0
C-1015               Nithin Kumar                                            0
C-1009               Sanjay Raj                                              0
C-1006               Kaviya Suresh                                           0
C-1013               Arjun Raj                                               0
C-1004               Anu Priya                                               2

CUSTOMER_ID          CUSTOMER_NAME                            NUMBER_OF_ORDERS
-------------------- ---------------------------------------- ----------------
C-1001               Beema ponting                                           2
C-1003               Arun kumar                                              2
C-1005               Reena das                                               2
C-1002               priya sakthi                                            2

15 rows selected.
```

4. Latest purchase details of customers.

```
SQL> SELECT
  2      C.*,
  3      O.ORDER_ID AS ORDER_ID,
  4      O.ORDER_DATE AS LATEST_PURCHASE_DATE,
  5      O.TOTAL_AMOUNT AS TOTAL_AMOUNT,
  6      O.ORDER_STATUS AS ORDER_STATUS
  7  FROM CUSTOMER C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  WHERE O.ORDER_DATE = (
 11      SELECT MAX(ORDERS.ORDER_DATE)
 12      FROM ORDERS
 13      WHERE ORDERS.CUSTOMER_ID = C.CUSTOMER_ID
 14  )
 15  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A   ORDER_ID LATEST_PU
-------------------------------- ---------- --------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
C-1001               Beema                ponting
beema200@edss.com               9234567856 14-FEB-23       1006 25-SEP-26
       55000 Delivered

C-1002               priya                sakthi
priyasakthi@edss.com            9856734456 23-SEP-24       1007 26-SEP-26
       67500 Pending

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A   ORDER_ID LATEST_PU
-------------------------------- ---------- --------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

C-1003               Arun                 kumar
arun200@edss.com                9876543212 15-OCT-24       1008 27-SEP-26
       45000 Shipped

C-1004               Anu                  Priya
anupriya200@edss.com            8765943256 23-MAY-25       1009 28-SEP-26

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A   ORDER_ID LATEST_PU
-------------------------------- ---------- --------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
        3400 Delivered

C-1005               Reena                das
reenadas200@edss.com            9087654567 30-APR-22       1011 30-SEP-26
       60000 Pending
```

6. Generate Multi-Table Business Reports :

Report 1: Customer Order Report:

Display:

- Customer name

- Order ID
- Order date
- Order status

```
SQL> SELECT
  2       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3       O.ORDER_ID,
  4       O.ORDER_DATE,
  5       O.ORDER_STATUS
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  ORDER BY O.ORDER_DATE;

CUSTOMER_NAME                                ORDER_ID ORDER_DAT
-------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Beema ponting                                      1001 20-SEP-26
Pending

priya sakthi                                       1002 21-SEP-26
Shipped

Arun kumar                                         1003 22-SEP-26
Delivered


CUSTOMER_NAME                                ORDER_ID ORDER_DAT
-------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Anu Priya                                          1004 23-SEP-26
Pending

Beema ponting                                      1006 25-SEP-26
Delivered

priya sakthi                                       1007 26-SEP-26
Pending


CUSTOMER_NAME                                ORDER_ID ORDER_DAT
-------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Arun kumar                                         1008 27-SEP-26
```

```
Anu Priya                                          1009 28-SEP-26
Delivered

Reena das                                          1010 29-SEP-26
Pending


CUSTOMER_NAME                                ORDER_ID ORDER_DAT
-------------------------------------------- ---------- ---------
ORDER_STATUS
--------------------
Reena das                                          1011 30-SEP-26
Pending


10 rows selected.
```

Report 2: Sales Report:

Display:

- Product name

● Quantity sold
● Total revenue

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCT P
  6  JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_REVENUE;

PRODUCT_NAME
--------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Fish tank
            2          6800

Unruled notes
            1          7500

Tiles
            1         55000


PRODUCT_NAME
--------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Television
            1         75000

Croptops and shirts
            2         90000

Laptops
            2        135000


6 rows selected.
```

Report 3: Payment Analysis Report:
Display:
● Payment mode
● Number of transactions

● Successful payments

```
SQL> SELECT
  2      PAYMENT.PAYMENT_MODE AS PAYMENT_MODE,
  3      COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  4      SUM(
  5          CASE
  6              WHEN PAYMENT.PAYMENT_STATUS = 'Completed' THEN 1
  7              ELSE 0
  8          END
  9      ) AS SUCCESSFUL_PAYMENTS
 10  FROM PAYMENT
 11  GROUP BY PAYMENT.PAYMENT_MODE
 12  ORDER BY NUMBER_OF_TRANSACTIONS;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- ------------------
Cash                                      2                  0
Card                                      3                  0
UPI                                       4                  0

SQL>
```

Report 4: Customer Purchase Analysis:

Find:

● Top purchasing customers

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID          CUSTOMER_NAME                          TOTAL_SPENDING
-------------------- -------------------------------------- --------------
C-1001               Beema ponting                                  125000
C-1002               priya sakthi                                   112500
C-1004               Anu Priya                                       78400
C-1005               Reena das                                       67500
C-1003               Arun kumar                                      48400
```

● Customers with maximum orders

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDER_ID) = (
 13      SELECT MAX(ORDER_COUNT)
 14      FROM (
 15          SELECT COUNT(ORDER_ID) AS ORDER_COUNT
 16          FROM ORDERS
 17          GROUP BY CUSTOMER_ID
 18      )
 19  );

CUSTOMER_ID          CUSTOMER_NAME                              NUMBER_OF_ORDERS
-------------------- ---------------------------------------- ----------------
C-1002               priya sakthi                                            2
C-1003               Arun kumar                                              2
C-1004               Anu Priya                                               2
C-1005               Reena das                                               2
C-1001               Beema ponting                                           2

SQL>
```

- Customers with highest spending

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTAL_AMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMER_ID
 18      )
 19  );

CUSTOMER_ID          CUSTOMER_NAME                            TOTAL_SPENDING
-------------------- ---------------------------------------- --------------
C-1001               Beema ponting                                    125000
```